

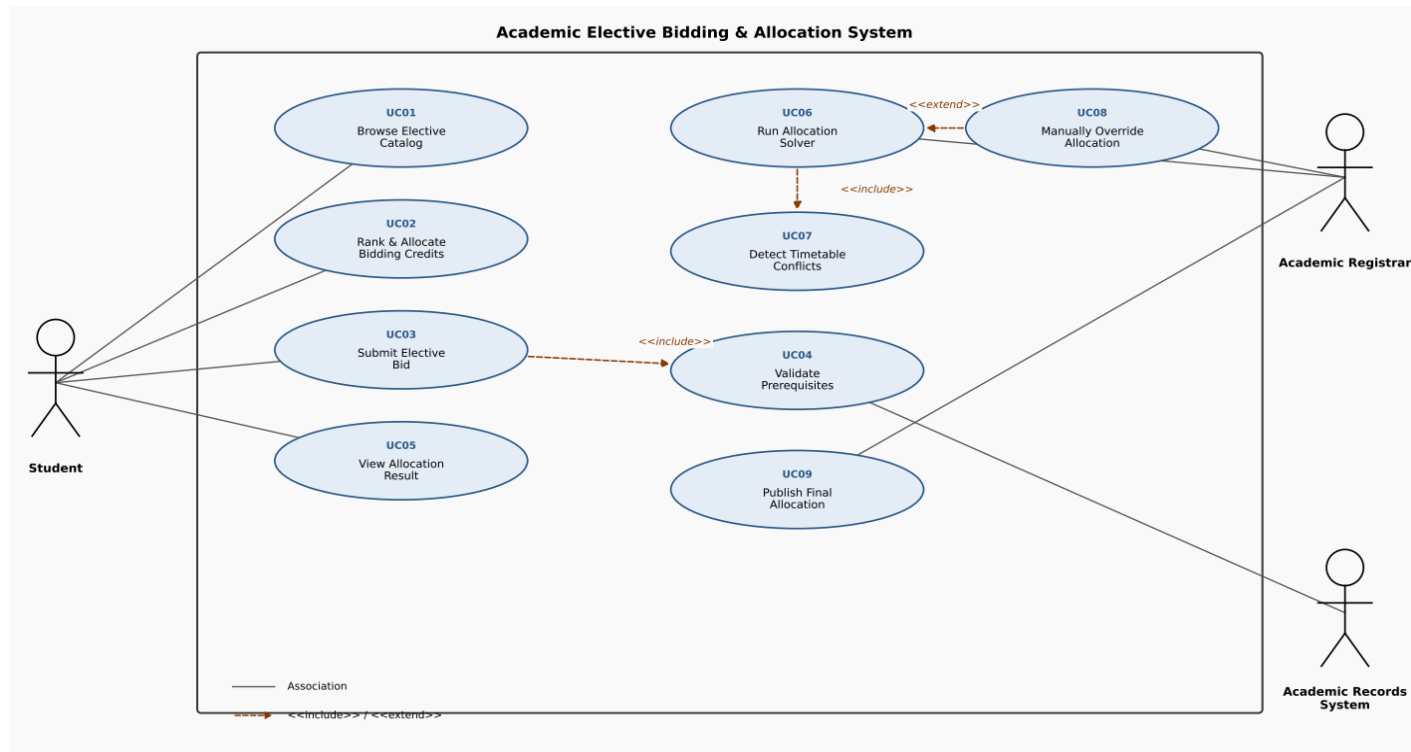
Academic Elective Bidding & Allocation System

Requirements Table — Lab 1: Requirements Engineering & UML Use-Case Modelling

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions before accepting a bid.	High	Pass: Credits deduct accurately and prerequisites are validated for every submitted bid. Fail: Total allocated credits exceed 100, or a bid with an unmet prerequisite is accepted.	Core mechanism that lets students express weighted preferences while protecting academic eligibility rules.
FR-002	Functional	The system shall accept elective bid submissions only within the registrar-configured bidding window and shall reject any submission attempted before it opens or after it closes.	High	Pass: Submission attempts outside the configured window are rejected with a clear error message and no credits are deducted. Fail: A bid is accepted, or credits are deducted, outside the configured bidding window.	Preserves fairness and a consistent, auditable timeline for all students competing for the same seats.
FR-003	Functional	The system shall check each ranked elective preference against the student's existing timetable and flag any preference whose session timing overlaps with an already-enrolled course.	High	Pass: Overlapping electives are flagged before final submission and excluded from allocation eligibility. Fail: A student is allocated two electives whose scheduled sessions overlap.	Prevents unusable allocations and reduces manual timetable-correction effort for the Registrar.
FR-004	Functional	The system shall notify each student of their final allocated elective(s) or waitlist status via the student portal and registered email within 2 hours of the allocation run completing.	Medium	Pass: 100% of students with a submitted bid receive a portal and email notification within 2 hours of allocation completion. Fail: Any student misses notification of results, or is notified after the 2-hour SLA.	Keeps students informed promptly so they can act on waitlist offers or add/drop windows without delay.
FR-005	Functional	The system shall allow the Academic Registrar to manually reassign or override a student's automated allocation for a specific case (e.g., medical or disability accommodation), and shall require a recorded justification note.	Medium	Pass: Every manual override is saved with the Registrar's ID, a timestamp, and a justification, and is reflected in the student's final allocation. Fail: An override is applied without a logged justification, or the student's final record is not updated.	Supports legitimate exceptions to the automated process while preserving accountability and an audit trail.

NFR-001	Nonfunctional	[Type: Performance & Security] The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load. Fail: The solver exceeds 30 seconds for a 5,000-bid batch, or fails security checks under load.	The allocation run is a single time-boxed batch job; slow processing delays results for the entire student cohort.
NFR-002	Nonfunctional	[Type: Reliability & Availability] The elective bidding portal shall maintain at least 99.5% uptime and support at least 5,000 concurrent student sessions with response times under 3 seconds during the published bidding window.	High	Pass: Load testing confirms $\geq 99.5\%$ uptime and $< 3s$ response time at 5,000 concurrent sessions across the full bidding window. Fail: Downtime exceeds the 0.5% budget, or response times exceed 3 seconds under peak concurrent load.	Bidding windows are strictly time-boxed; an outage or slowdown can unfairly disadvantage students who cannot resubmit in time.

UML Use-Case Diagram



Use-Case Flow Specification

Academic Elective Bidding & Allocation System — UC-03: Submit Elective Bid

Use Case ID: UC-03	Use Case Name: Submit Elective Bid
Primary Actor: Student	Secondary Actors: Academic Records System
Related Requirements: FR-001, FR-002, FR-003	Trigger: Student selects “Bid for Electives” during an open bidding window

Preconditions :

- The student is authenticated and enrolled in a program with elective slots available for the current term.
- The elective bidding window configured by the Academic Registrar is currently open (see FR-002).
- The student holds an available balance of 100 bidding credits that has not yet been spent.
- The elective catalog, including seat capacity, prerequisites, and session timings, has been published.

Postconditions :

Success guarantee:

- The student's ranked preferences and credit allocation are stored as a timestamped bid record with status “Submitted.”
- The allocated credits are deducted from the student's available balance.
- Every selected elective has passed prerequisite validation and timetable-collision screening.

Minimal guarantee (failure):

- No partial bid is saved; the student's credit balance and prior bid state remain unchanged.

Main Success Scenario :

- The student selects “Bid for Electives” from the student portal.
- The system displays the list of open electives with seat capacity, prerequisites, and session timing.
- The student ranks the desired electives in order of preference.
- The student allocates bidding credits (total ≤ 100) across the ranked preferences.
- The system validates that the total allocated credits do not exceed 100.
- The system validates prerequisite completion for each selected elective by querying the Academic Records System (<<include>> Validate Prerequisites, FR-001).
- The system checks each preference against the student's existing timetable for session overlaps (FR-003).
- The student reviews the validated bid summary and confirms submission.
- The system timestamps and stores the bid, and deducts the allocated credits from the student's balance.
- The system displays a confirmation message with a bid reference number. Use case ends successfully.

Alternate Flow A — Credits Exceed 100 (from Step 5) :

- The system displays an error indicating the total exceeds the 100-credit limit.
- The student is prompted to revise the credit allocation.
- The student resubmits an allocation within the limit; the flow resumes at Step 5.

Alternate Flow B — Prerequisite Not Met (from Step 6) :

- The system flags the specific elective(s) for which a prerequisite is unmet.
- The student is prompted to remove or replace the flagged elective before proceeding.
- Once the flagged elective is resolved, the flow resumes at Step 7.